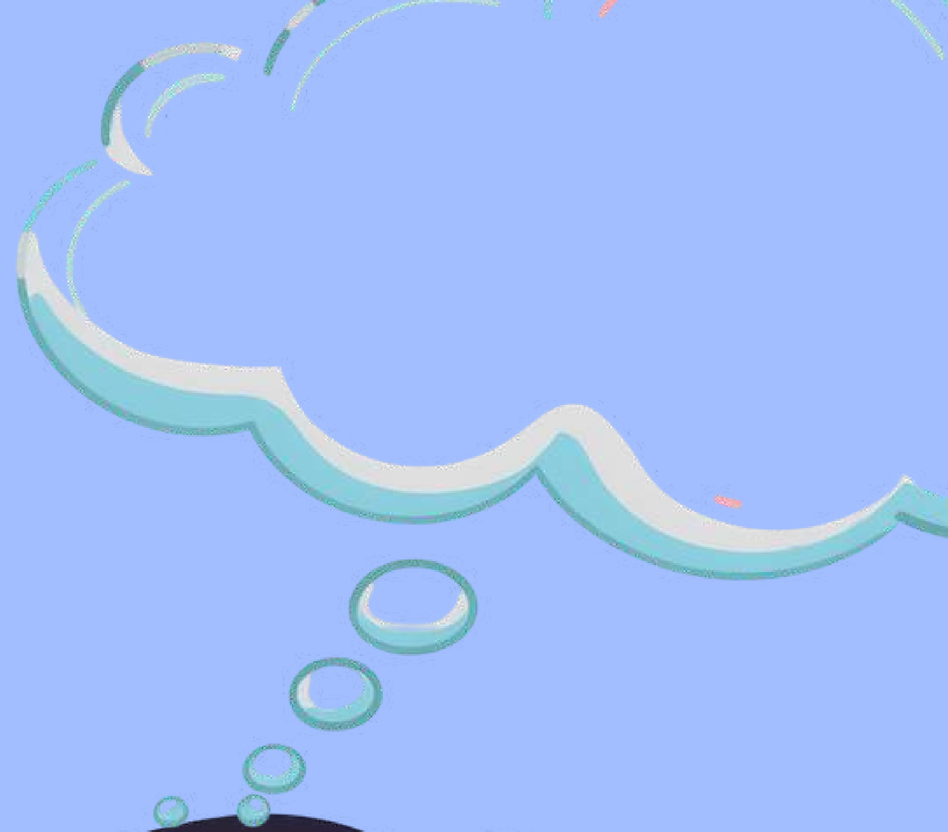
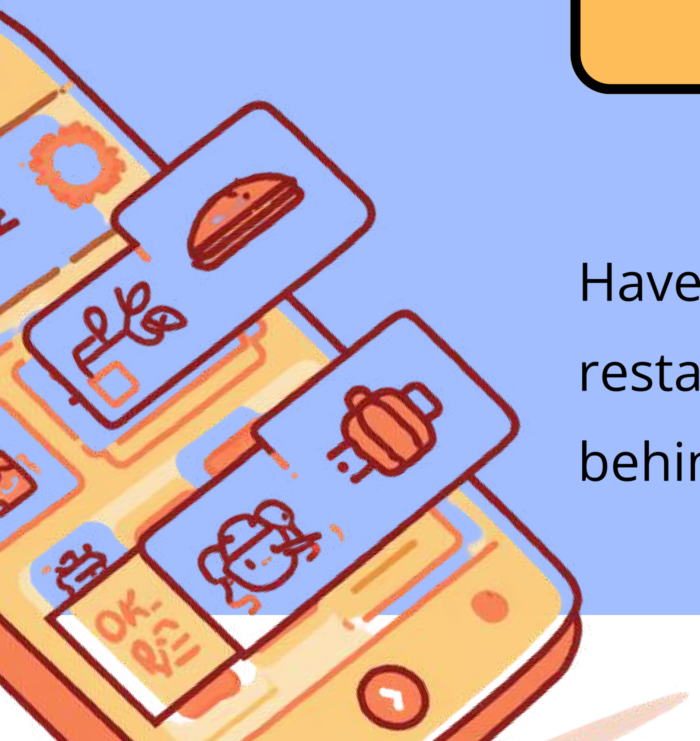
DIGITAL WAITERS: INTRODUCTION TO REST APIS



HOW DOES ZOMATO KNOW WHICH RESTAURANTS ARE OPEN?

Think About It: Real-Time Restaurant Info

The Secret? Something Called an API!



Have you ever wondered how the app knows exactly which restaurants are open right now? Let's discover the magic behind the scenes!

REST API Design

WHAT IS AN API? THE DIGITAL WAITER ANALOGY

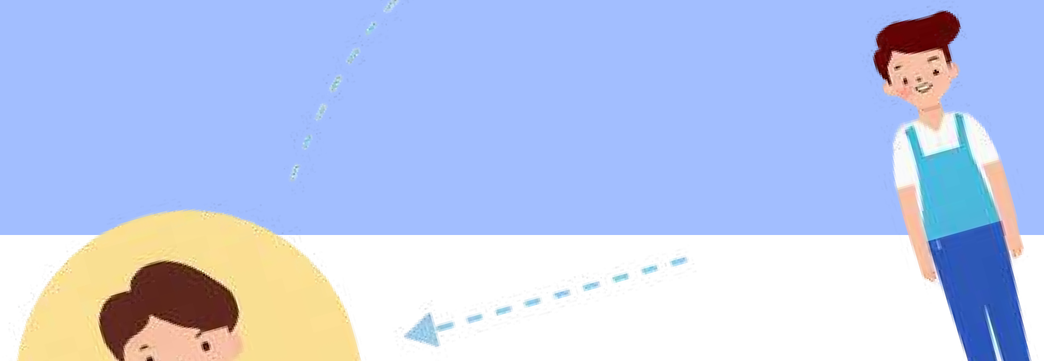
You (Customer) tell the Waiter (API) what you want

Waiter takes your order to the Kitchen (Server)

Waiter brings back your food (Response)

API connects you to the data you need!

Just like ordering at a restaurant!



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WHAT IS AN ENDPOINT?

The Counter Analogy – Like Big Bazaar!

Think of a big store like Big Bazaar or Reliance Smart. There are different counters: Milk Counter, Electronics Counter, Grocery Counter. Each counter gives you specific items!

Each Counter = A Different Endpoint

In APIs, endpoints work the same way! /milk gives you milk info, /electronics gives you gadget info. Each endpoint serves a specific purpose!

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JSON: THE COMMON LANGUAGE FOR COMPUTERS

JSON = JavaScript Object Notation

It Helps Computers Understand Each Other

Just like we use Hindi or English to talk to each other, computers use JSON to communicate. It's a simple, organized way for computers to share information clearly!

THE 4 BASIC API ACTIONS

GET: Read Information

Ask for today's menu from the waiter

PUT: Update Info

Change your order before it's ready

POST: Create New Info

Place your order at the canteen

DELETE: Remove Info

Cancel your order completely

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HOW THE REQUEST-RESPONSE CYCLE WORKS

Step 1: You Make a Request

Step 2: Waiter Takes It to Kitchen

Step 3: Kitchen Prepares Response

Step 4: Waiter Brings Back Your Food

Step 5: You Receive the Response!

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ACTIVITY TIME: DESIGN YOUR SCHOOL CANTEEN API

Create Your Blueprint

Design your own API! Think of what your canteen needs and create a blueprint

Name Your Endpoints

Pick names like /canteen/menu or /canteen/order for different services

Choose Your Actions

Decide if each endpoint uses GET, POST, PUT, or DELETE actions

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SCHOOL CANTEEN API ENDPOINTS

/canteen/menu – GET – Returns Snacks & Prices

Ask for the menu and see all available snacks like samosas, sandwiches, and juice with their prices. The server sends back a list!

/canteen/order – POST – Sends Your Order

Place your snack order by sending what you want. The kitchen receives it and prepares your food!

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LET'S SEE A VISUAL DEMO!

Building Your First API Endpoint

See How Easy It Is to Create!

We'll use a mock API tool to build a simple "Hello World" endpoint. Watch how we set up the endpoint and see the magic happen!

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DEMO RESULT: HELLO WORLD ENDPOINT RESPONSE

Our Endpoint Successfully Responds!

See the JSON Response Below

When we ask the endpoint, it says: "Hello, World!" This is the response we get back from our very first API endpoint!

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RECAP QUIZ: WHICH METHOD IS IT?

See Today's Menu

GET, POST, PUT, or DELETE?

Place a New Order

GET, POST, PUT, or DELETE?

Change Your Order

GET, POST, PUT, or DELETE?

Cancel Your Order

GET, POST, PUT, or DELETE?

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WHAT WE LEARNED TODAY

API is Like a Digital Waiter

Endpoints Are Specific Counters

Four Actions: GET, POST, PUT, DELETE

We Designed Our Own API Endpoints

Request-Response Cycle Works!

THANK YOU!

QUESTIONS?